Question ID: 356ad646

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Medium

Question

Baltimore, Maryland, has installed engineered structures along 71% of its shoreline to protect infrastructure from wave erosion and other hazards, a practice known as shoreline hardening. To evaluate the responses of waterbirds to two types of hardening structures—riprap and bulkheads—Diann Prosser et al. surveyed waterbird communities consisting of the tundra swan, the great blue heron, and 62 other species at different sites in the Chesapeake Bay on the US East Coast. Utilizing the Index of Waterbird Community Integrity (IWCI), on which a high score corresponds to high community integrity, the researchers found that bulkheads are more strongly negatively correlated with waterbird community integrity than is riprap.

Which finding, if true, would most directly illustrate the researchers' finding?

Answer

A. The difference in average IWCI scores for waterbird communities at Stony and Old Road, two sites with a higher percentage of shoreline consisting of bulkheads than of riprap, was statistically insignificant.

B. Waterbird communities at Old Road, a site with a relatively high percentage of shoreline consisting of bulkheads, had lower average IWCI scores than did waterbird communities at Miles, a site with a relatively high percentage of shoreline consisting of riprap.

C. Waterbird communities at Curtis, a site with a high percentage of shoreline consisting of bulkheads and riprap, had lower average IWCI scores than did waterbird communities at Onancock, a site with a low percentage of shoreline consisting of bulkheads and riprap.

D. Waterbird communities at Curtis, a site with equal percentages of shoreline consisting of bulkheads and riprap, had higher average IWCI scores than did waterbird communities at Miles, a site with different percentages of shoreline consisting of bulkheads and riprap.

Correct Answer: B

Rationale

Choice B is the best answer because it presents a finding that, if true, would most directly illustrate the researchers' finding about waterbird responses to shoreline hardening structures. The text explains that using the IWCI, an index on which higher scores indicate higher community integrity, researchers found that bulkheads are more strongly negatively correlated with waterbird community integrity than riprap is—that is, that bulkheads reduce the integrity of waterbird communities more than riprap does. The finding that waterbird communities at a site with a relatively high percentage of bulkheads along the shoreline (Old Road) had lower average IWCI scores than waterbird communities at a site with a relatively high percentage of riprap along the shoreline (Miles) did would illustrate the researchers' finding, since it would be an example of lower waterbird community integrity in a location with substantial bulkhead presence than in a location with substantial riprap presence.

Choice A is incorrect because a finding of similar waterbird community integrity in two sites predominated by bulkheads wouldn't reveal anything about waterbird community integrity at sites predominated by riprap or how waterbird community integrity compares between the two types of sites. Thus, the finding wouldn't illustrate the discovery of a difference in how bulkheads and riprap correlate with waterbird community integrity. Choice C is incorrect because it presents a finding that compares waterbird community integrity at sites with different overall amounts of shoreline hardening but without distinguishing between bulkheads and riprap. Such a distinction would be necessary to illustrate the researchers' finding of a difference in how bulkheads and riprap each correlate with waterbird community integrity. Choice D is incorrect because it presents a finding that addresses a difference in waterbird community integrity between one site with equal percentages of bulkheads and riprap along its shoreline (Curtis) and one site with unequal percentages of bulkheads and riprap (Miles) but without indicating whether bulkheads or riprap were more prevalent at Miles. This finding wouldn't provide any clear information about how bulkheads and riprap each correlate with waterbird community integrity, so it wouldn't illustrate the researchers' finding of a difference between those correlations.